Charles-Gérard LUCAS

French citizen, born on October 6th 1995

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Research experience

Oct. 2025 - now	**Post-doctoral researcher**, *Centre INRIA de l'Université Grenoble-Alpes*, Grenoble, France.
Activity	Modeling high-contrast intensity observations for the study of circumstellar disks.
Supervisors	Julien Mairal, Olivier Flasseur and Jean Ponce.
Dec. 2023 - May 2025	**Post-doctoral researcher**, *Department of Mathematics & Statistics, San Diego State University*, San Diego, California, USA.
Activity	*Empirical wavelet transform: theoretical developments and applications.*
Supervisor	Jérôme Gilles.
Oct. 2020 - Oct. 2023	**PhD Thesis in Physics**, *Physics Laboratory, École Normale Supérieure de Lyon*, Lyon, France.
Title	Multivariate self-similarity: estimation of the self-similarity exponents, bootstrap test for the equality of exponents and applications
Supervisors	Patrice Abry and Herwig Wendt.
May 2020 - Sept. 2020	**Master 2 Internship**, *Physics Laboratory, École Normale Supérieure de Lyon*, Lyon, France.
Supervisors	Nelly Pustelnik, Barbara Pascal and Patrice Abry.
Title	*Contour detection using Mumford-Shah.*
Keywords	Image processing, inverse problem, proximal algorithms, risk estimator.
Apr. 2019 - Sept. 2019	**Master 2 Internship**, *Cosmology and Statistics Laboratory (CosmoStat), CEA Saclay*, Gif-sur-Yvette, France.
Supervisors	Morgan Schmitz and Jean-Luc Starck.
Title	*Point Spread Function modeling in astronomy.*
Keywords	Image processing, inverse problem, proximal algorithms, wavelets.
Oct. 2018 - Mar. 2019	**Master 2 Project**, *Laboratory of Medical Information Processing (LaTIM), IMT Atlantique*, Brest, France.
Supervisor	Chafiaa Hamitouche.
Title	*Analysis of morpho-functional signatures obtained using dual quaternions on a group of patients who are candidates for a knee arthroplasty.*
Keywords	Morpho-functional modeling, dual quaternions, Lie algebra, dynamic time warping.

Jul. 2018	**Bachelor Internship**, *Laboratory of Medical Information Processing (LaTIM), IMT Atlantique*, Brest, France.
Supervisor	John Puentes.
Title	*Segmentation of megakaryocytes in biopsy images.*
Keywords	Image processing, pattern recognition.

Education

2020 - 2023 **PhD Thesis in Physics, specialty Signal and Image Processing**, *École Normale Supérieure de Lyon*, Lyon, France.

2019 - 2020 **Master 2 Mathematics and applications, course Optimization**, *Université Paris-Saclay*, Palaiseau, France. Equivalent to a Master of Science degree in Applied Mathematics.

2018 - 2019 **Master 2 Signal, Image, Systems, Automatic (SISEA), course Image processing**, *Université de Rennes 1*, Rennes, France. Equivalent to a Master of Science degree in Image Processing.

2016 - 2019 **Diplôme d'Ingénieur Généraliste**, *IMT Atlantique*, Brest, France. Equivalent to Master of Science degree in Telecommunication Engineering.

2016 - 2018 **Licence & Master 1 Mathématiques Fondamentales**, *Université de Bretagne Occidentale*, Brest, France.
Equivalent to a Bachelor of Science degree in Fundamental Mathematics.

2013 - 2016 **Classe Préparatoire MP**, *Lycée Jeanne d'Albret*, Saint-Germain-en-Laye, France. Intensive preparation for the national competitive entrance examination to leading French Engineering Schools, specializing in Mathematics and Physics.

Publications & communications

All papers are available on HAL: `https://cv.hal.science/charles-gerardlucas`.

In preparation

2. **Charles-Gérard Lucas**, Patrice Abry, Herwig Wendt, and Gustavo Didier. Learning graph structures for multivariate self-similar systems using a wavelet-domain bootstrap testing procedure. In preparation

1. **Charles-Gérard Lucas**, Olivier Flasseur, Julien Mairal, and Jean Ponce. Multi-distribution variational reconstruction of extended circumstellar environments in high-contrast direct imaging. In preparation

International journals

5. **Charles-Gérard Lucas** and Jérôme Gilles. Multidimensional empirical wavelet transform. *SIAM Journal on Imaging Sciences*, 18(2):906–935, 2025

4. **Charles-Gérard Lucas** and Jérôme Gilles. Demon registration for 2D empirical wavelet transforms. *Foundations*, 4(4):690–703, 2024

3. **Charles-Gérard Lucas**, Gustavo Didier, Herwig Wendt, and Patrice Abry. Multivariate selfsimilarity: Multiscale eigenstructures for selfsimilarity parameter estimation. *IEEE Transactions on Signal Processing*, 72:1739–1749, 2024

2. **Charles-Gérard Lucas**, Barbara Pascal, Nelly Pustelnik, and Patrice Abry. Hyperparameter selection for Discrete Mumford–Shah. *Signal, Image and Video Processing*, 17(5):1897–1904, 2023

1. Patrice Abry, Nelly Pustelnik, Stéphane Roux, Pablo Jensen, Patrick Flandrin, Rémi Gribonval, **Charles-Gérard Lucas**, Éric Guichard, Pierre Borgnat, and Nicolas Garnier. Spatial and temporal regularization to estimate COVID-19 reproduction number R(t): Promoting piecewise smoothness via convex optimization. *PLoS ONE*, 15(8):e0237901, 2020

International conference proceedings

5. **Charles-Gérard Lucas**, Herwig Wendt, Patrice Abry, and Gustavo Didier. Counting distinct multivariate self-similarity parameters using a bootstrap-driven graph clustering approach. In *2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 826–830, Barcelona, Spain, 2026. IEEE. Poster presentation.

4. **Charles-Gérard Lucas**, Patrice Abry, Herwig Wendt, and Gustavo Didier. Epileptic seizure prediction from eigen-wavelet multivariate selfsimilarity analysis of multi-channel EEG signals. In *2023 31st European Signal Processing Conference (EUSIPCO)*, pages 970–974, Helsinki, Finland, 2023. IEEE. Poster presentation.

3. **Charles-Gérard Lucas**, Patrice Abry, Herwig Wendt, and Gustavo Didier. Drowsiness detection from polysomnographic data using multivariate selfsimilarity and eigen-wavelet analysis. In *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, pages 2949–2952, Glasgow, Scotland, 2022. IEEE. Oral presentation.

2. **Charles-Gérard Lucas**, Patrice Abry, Herwig Wendt, and Gustavo Didier. Counting the number of different scaling exponents in multivariate scale-free dynamics: Clustering by bootstrap in the wavelet domain. In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5513–5517, Singapore, 2022. IEEE. Oral and poster presentations (online).

1. **Charles-Gérard Lucas**, Patrice Abry, Herwig Wendt, and Gustavo Didier. Bootstrap for testing the equality of selfsimilarity exponents across multivariate time series. In *2021 29th European Signal Processing Conference (EUSIPCO)*, pages 1960–1964, Dublin, Ireland, 2021. IEEE. Oral presentation (online).

National conference proceedings

2. **Charles-Gérard Lucas**, Patrice Abry, Herwig Wendt, Gustavo Didier, and Oliver Orejola. Bootstrap based test for the unimodality of estimated Hurst exponents. Performance assessment in a high-dimensional analysis setting. In *XXIXème Colloque Francophone de Traitement du Signal et des Images (GRETSI 2023)*, Grenoble, France, 2023. Poster presentation.

1. **Charles-Gérard Lucas**, Herwig Wendt, Patrice Abry, and Gustavo Didier. Multivariate time-scale bootstrap for testing the equality of selfsimilarity parameters. In *XXVIIIème Colloque Francophone de Traitement du Signal et des Images (GRETSI 2022)*, Nancy, France, 2022. Poster presentation.

Seminar talks

16. Annual Symposium of the GdR MaDICS
Reconstruction of circumstellar environments in high-contrast direct imaging
Avignon, France, June 2nd 2026 (poster)

15. Thoth seminar, INRIA Grenoble
Multivariate self-similarity: estimation of self-similarity exponents and biomedical application
Montbonnot-Saint-Martin, France, January 27th 2026

14. Computer Science, Signals and Systems Laboratory of Sophia Antipolis (I3S)
Multivariate self-similarity: estimation of self-similarity exponents and biomedical application
Sophia-Antipolis, France, December 16th 2025

13. GIPSA-lab
Kernel-based empirical wavelet transform for adaptive time-frequency representation
Saint-Martin-d'Hères, France, November 27th 2025

12. Jean Kuntzmann Laboratory
A kernel-based empirical wavelet transform for adaptive time-frequency representation
Saint-Martin-d'Hères, France, November 24th 2025

11. Laboratory of Signals and Systems (L2S)
Kernel-based empirical wavelet transform: properties, implementation and applications
Gif-sur-Yvette, France, November 7th 2025 (online)

10. Weierstrass Institute for Applied Analysis and Stochastics
Multidimensional kernel-based empirical wavelet transform
Berlin, Germany, July 1st 2025 (online)

9. Biomedical and Astronomical Signal Processing Laboratory
Multidimensional kernel-based empirical wavelet transform
Edinburgh, United Kingdom, May 23rd 2025 (online)

8. Research Seminars: Probability and Statistics, Tulane University
Multidimensional kernel-based empirical wavelet transform
New Orleans, Louisiana, USA, February 12th 2025

7. Laboratory of the Integration from Material to System (IMS)
Multidimensional empirical wavelet transform
Bordeaux, France, December 4th 2024 (online)

6. Marseille Institute of Mathematics
Multivariate self-similarity: modeling, parameter estimation and application
Marseille, France, October 11th 2024 (online)

5. Computational Science Research Center, San Diego State University
   Multivariate self-similarity: estimation of the self-similarity exponents and application
   San Diego, California, USA, September 19th 2024

4. FracText, Institute of the Anglophone, Germanic, and Romanic Worlds (IMAGER)
   Identification of the original or translated nature of a text
   Créteil, France, September 29th-30th 2022

3. PhD Day of the Physics Laboratory, ENS Lyon
   Testing pairwise equality of multivariate self-similarity exponents
   Lyon, France, June 22nd 2022

2. GDR AMA Days - CNRS
   Clustering self-similarity exponents of multivariate time series by a wavelet-domain bootstrap
   Porquerolles, France, September 27th-30th 2021

1. PhD Day of the Physics Laboratory, ENS Lyon
   Multivariate self-similarity exponent estimation
   Lyon, France, May 18th 2021

Summer school

1. Harmonic and Multifractal Analyses: from Mathematics to Quantitative Neurosciences
   Participation, Montréal, Canada, July 3rd-14th 2023

Softwares

6. DSM-REX
   Python toolbox for the reconstruction of circumstellar disks with smooth and sparse regularizations, using deep learning (denoising score matching) of the nuisance model (due to the host star).

5. MDREX (`github.com/charlesglucas/MDREX`)
   Python toolbox for the reconstruction of circumstellar disks with smooth and sparse regularizations, with estimation of the parameters of the Gaussian nuisance model (due to the host star).

4. Contribution to Empirical Wavelets (`github.com/jegilles/Empirical-Wavelets`)
   Matlab toolbox for the 2D empirical wavelet transform from Gabor and Shannon wavelet kernels using mapping estimation.

3. OFBM Tools (`github.com/charlesglucas/ofbm_tools`)
   Matlab toolbox for operator fractional Brownian motion (ofBm) analysis. Estimation and counting of scaling parameters of multivariate self-similar signals.

2. SUGAR D-MS (`github.com/charlesglucas/sugar_dms`)
   Matlab toolbox for joint denoising and contour detection of images. Minimization of the Discrete Mumford-Shah functional with automatic selection of hyperparameters.

1. RCA (`github.com/charlesglucas/rca`)
   Python toolbox for both Point Spread Function (PSF) estimation and galaxy image deconvolution simultaneously using star images.

Internship supervision

Sept. 2024 - May 2025	**Master thesis** of Sam Persaud, *Department of Mathematics, San Diego State University*, San Diego, California, USA, co-supervised with Jérôme Gilles.
Topic	Study of the hyperparameters of the Demons functional for homeomorphism estimation.

Commitments in the Scientific Community

5. **Volunteer** for international conference EUSIPCO organization — 26 - 30 aug. 2024
   Organized by Patrice Abry and Maria Sabrina Greco in Lyon, France.

4. **Co-organizer** of PhD student seminars — nov. 2022 - june 2023
   Physics Laboratory, ENS Lyon, France

3. **PhD student representative** — oct. 2022 - sept. 2023
   Physics Laboratory, ENS Lyon, France

2. **Volunteer** for international conference CCS organization — 25 - 29 oct. 2021
   Organized by Pierre Borgnat and Márton Karsai in Lyon, France.

1. **Co-organizer** of the PhD Day — 22 june 2022
   Physics Laboratory, ENS Lyon, France
   Day of research talks of first and second year PhD students in Physics.

Teaching experience

Université Grenoble-Alpes

Bachelor of Physics, Chemistry, Mechanics and Mathematics

- Real analysis - First year (22h) — 2025 - 2026
  Practical exercices and written examinations
  Functions of a real variable, antiderivatives and integrals, linear differential equations.
  Course coordinator: Faouzi Triki

San Diego State University

Bachelor

- Advanced Calculus I (26h) — 2024 - 2025
  Lectures, practical exercices and written examinations
  Real numbers, sequences and limits, limits of functions of a real variable, continuity of functions, derivatives.

École Normale Supérieure (ENS) de Lyon

Master of Complex Systems

- Complex networks - Second year (12h) 2021 - 2022, 2022 - 2023
  Practical exercices and numerical implementation (Python)
  Fundamentals of Network Science, e.g., classic random models, centralities, small-world phenomenon ; Advances topics, e.g., dynamic networks, graph algorithmic, community detection, machine learning on graphs.

École supérieure de chimie, physique, électronique de Lyon (CPE Lyon)

Master of Chemical Engineering

- Random Signal Processing - First year (16h) 2022 - 2023
  *Practical exercices and numerical implementation (*Matlab*)*
  Random signals, spectral estimation, quadratic detection, linear prediction.

Université Claude Bernard Lyon 1

Bachelor of Mathematics

- Introduction to numerical analysis - Second year (12h) 2021 - 2022, 2022 - 2023
  Practical exercices, numerical implementation (Python) and written examinations
  Polynomial interpolation, quadrature method, root-finding algorithms, numerical methods for differential equations.

- Geometric algebra - Second year (40h) 2021 - 2022
  Practical exercices and written examinations
  Inner product, orthogonality, orthogonal projection on finite-dimensional subspaces, affine hyperplane in Euclidean spaces, vectorial isometry in Euclidean spaces, vectorial endomorphism in Euclidean spaces.

- Linear and bilinear algebra, matrix analysis - Third year (12h) 2021 - 2022
  Practical exercices
  Quadratic forms, endomorphism in Euclidean space, endomorphism in Hermitian space, linear systems.

- Fundamentals of mathematics - First year (24h) 2020 - 2021
  Colles (oral examinations)
  Complex numbers, sequences and limits, real-valued functions of a real variable, limits and continuity, derivation of real-valued functions, integer arithmetic, polynomials.

- Basic mathematical techniques - First year (40h) 2020 - 2021
  Lectures, practical exercices and written examinations
  Riemann integration, first and second order linear differential equations, complex numbers, vector spaces, geometry in the plane and in space.

Skills

Computing	MATLAB, Python, LaTeX
Graphics editor	Inkscape
Languages	French (native), English (advanced), Spanish (advanced)